

**NICKEL****ICSC: 0062 (April 2017)**

Metallic nickel

**CAS #: 7440-02-0****EC Number: 231-111-4**

|                             | <b>ACUTE HAZARDS</b>                                                                                                                    | <b>PREVENTION</b>                                                                                  | <b>FIRE FIGHTING</b>                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>FIRE &amp; EXPLOSION</b> | Flammable as dust. Gives off irritating or toxic fumes (or gases) in a fire. Finely dispersed particles form explosive mixtures in air. | Closed system, dust explosion-proof electrical equipment and lighting. Prevent deposition of dust. | Use dry sand, dry powder. NO carbon dioxide. NO water. |

| <b>PREVENT DISPERSION OF DUST! AVOID ALL CONTACT!</b> |                             |                                                                                    |                                                                                                                                            |
|-------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <b>SYMPTOMS</b>             | <b>PREVENTION</b>                                                                  | <b>FIRST AID</b>                                                                                                                           |
| <b>Inhalation</b>                                     | Cough. Shortness of breath. | Use local exhaust or breathing protection.                                         | Fresh air, rest.                                                                                                                           |
| <b>Skin</b>                                           | Redness.                    | Protective gloves. Protective clothing.                                            | Wear protective gloves when administering first aid. Remove contaminated clothes. See Notes. Rinse and then wash skin with water and soap. |
| <b>Eyes</b>                                           | Redness.                    | Wear safety spectacles or eye protection in combination with breathing protection. | First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.         |
| <b>Ingestion</b>                                      |                             | Do not eat, drink, or smoke during work.                                           | Rinse mouth.                                                                                                                               |

| <b>SPILLAGE DISPOSAL</b>                                                                                                                                                                                                                                                                                                                              | <b>CLASSIFICATION &amp; LABELLING</b>                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personal protection: particulate filter respirator adapted to the airborne concentration of the substance. Do NOT let this chemical enter the environment. Sweep spilled substance into sealable containers. If appropriate, moisten first to prevent dusting. Carefully collect remainder. Then store and dispose of according to local regulations. | <b>According to UN GHS Criteria</b><br><br><br><b>DANGER</b><br>May cause an allergic skin reaction<br>May cause allergy or asthma symptoms or breathing difficulties if inhaled<br>Suspected of causing cancer if inhaled<br>Causes damage to the lungs through prolonged or repeated exposure if inhaled<br>Harmful to aquatic life with long lasting effects |
| <b>STORAGE</b>                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Store only in original packaging. Cool. Well closed. Separated from strong oxidants and acids. Store in an area without drain or sewer access.                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>PACKAGING</b>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                                                                                                                                                       | <b>Transportation</b><br><b>UN Classification</b>                                                                                                                                                                                                                                                                                                                                                                                                    |

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**NICKEL****ICSC: 0062****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

SILVERY METALLIC LUSTROUS SOLID IN VARIOUS FORMS.

**Physical dangers**

If dry, it can be charged electrostatically by swirling, pneumatic transport, pouring, etc. Dust explosion possible if in powder or granular form, mixed with air.

**Chemical dangers**

Reacts violently with acids. This produces flammable hydrogen. This generates fire and explosion hazard. Reacts violently with strong oxidants. This generates fire and explosion hazard. This produces toxic fumes of nickel monoxide. See ICSC 0926. On combustion, forms toxic gases and vapours including nickel (II) oxide (see ICSC 0926) and nickel carbonyl (see ICSC 0064).

Formula: Ni

Atomic mass: 58.7

Boiling point: 2730°C

Melting point: 1455°C

Density: 8.9 g/cm<sup>3</sup>

Solubility in water, mg/l at 37°C: 1.1 (practically insoluble)

**EXPOSURE & HEALTH EFFECTS****Routes of exposure**

The substance can be absorbed into the body by inhalation of dust.

**Effects of short-term exposure**

May cause mechanical irritation. Inhalation of fume may cause pneumonitis.

**Inhalation risk**

Evaporation at 20°C is negligible; a harmful concentration of airborne particles can, however, be reached quickly when dispersed.

**Effects of long-term or repeated exposure**

Repeated or prolonged contact may cause skin sensitization. Repeated or prolonged inhalation may cause asthma. The substance may have effects on the respiratory tract. This may result in chronic inflammation of the respiratory tract and fibrosis. This substance is possibly carcinogenic to humans if inhaled.

**OCCUPATIONAL EXPOSURE LIMITS**

TLV: (inhalable fraction): 1.5 mg/m<sup>3</sup>, as TWA; A5 (not suspected as a human carcinogen); BEI issued.

MAK: (inhalable fraction): sensitization of respiratory tract and skin (SAH); carcinogen category: 1

**ENVIRONMENT**

The substance is harmful to aquatic organisms. The substance may cause long-term effects in the aquatic environment.

**NOTES**

At high temperatures, toxic fumes of nickel(II)oxide may be formed (see ICSC 0926).

Depending on the degree of exposure, periodic medical examination is suggested.

The symptoms of asthma often do not become manifest until a few hours have passed and they are aggravated by physical effort.

Anyone who has shown symptoms of sensitization due to this substance should avoid all further contact with nickel, nickel compounds and other metal compounds of e.g. copper, chromium and cobalt.

Isolate contaminated clothing by sealing in a bag or other container.

Do NOT take working clothes home.

**ADDITIONAL INFORMATION****EC Classification**

Symbol: Xn; R: 40-43; S: (2)-22-36

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